Diagnosis : the future

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DIAGNOSIS: THE FUTURE

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Control of human African trypanosomiasis (HAT) actually depends on a combination of case detection, curative treatment, vector control (plus treatment of the animal reservoir for *T.b. rhodesiense*). Correct diagnosis therefore plays a pivotal role in control of the disease. At present diagnosis of trypanosomiasis infection follows a multi-step pathway: screening by serological tests (not done for *T.b. rhodesiense*), diagnostic confirmation by parasitological confirmation and stage determination to assess central nervous system involvement. After treatment, a patient should be followed up for 2 years to confirm cure. In each of these steps, one is confronted with the shortcomings of the actual diagnostic and staging techniques.

The common test for screening the population at risk for *T.b. gambiense* infection is the card agglutination test for trypanosomiasis (CATT/*T.b. gambiense* or CATT). CATT is effective, and detection of antibodies against trypanosomes using CATT is a sensitive indicator of infection. The test specificity is high, around 95%. However, the positive predictive value of the CATT strongly decreases when the prevalence of the disease decreases (with 95% specificity, at prevalences of respectively 5, 2 and 0.5%, positive predictive values decrease from 50% to 28% to 9%). Since the test is used for mass screening in populations where the prevalence of HAT can be low (<2%), a large number of positive results are false-positives. Other limitations are: i) CATT is not applicable for *T.b. rhodesiense* infections; ii) CATT has lower sensitivity in some foci in Central and West Africa; iii) CATT can only be performed after a basic training; iv) there is need for a cold chain (4°C) for storage of CATT reagents; v) performance of the CATT test requires power supply to make the agitator run (220 V or car battery); vi) CATT is incapable of differentiating between active and cured infection, as antibodies tend to stay in the blood for prolonged periods after curative treatment; vii) CATT is manufactured using whole *T.b. gambiense* parasites recovered from infected laboratory animals. The production process is complex, laborious and dangerous; viii) CATT is produced in 50-test packages, and is not yet available as an individual test. This contributes to higher costs and wastage if the test is used for individual diagnosis.

Tests with higher specificity are therefore needed. Such tests could also eliminate the need for parasitological confirmation. A simple serological test for *T.b. rhodesiense* might allow more efficient mass screening for east-African trypanosomiasis in high risk areas and would result in less parasitological examinations, thus alleviating the work load. Moreover, relying less on clinical symptoms, may lead to detection of more early stage cases which might remain undetected at present.

Microscopy is laborious and requires highly skilled technicians. The best techniques, relying on concentration of trypanosomes, are costly, require equipment such as centrifuges and depend on electrical power. Sensitivity at its best remains limited to about 70-75% [1]. As a consequence of the limited sensitivity of parasite detection in combination with the sometimes low parasitemia, patients may go undiagnosed [2]. Molecular diagnostics have remained impractical and impossible to perform in the endemic rural areas due to absence of a developed infrastructure and skilled personnel.

Individuals in whom infection is confirmed undergo lumbar puncture to determine whether specific treatment for late stage disease is required. Lumbar puncture is invasive and painful. Trypanosomes in cerebrospinal fluid (CSF) are often scanty and reliability of the white blood cell count is limited, especially for relatively normal cell counts. Moreover, actual cut-off values are controversial.

The therapeutic outcome is monitored during a follow-up period of 2 years during which patients should come back every 6 months. This long follow-up period poses several problems. Relapses often are detected only in CSF [3], and parasites may remain undetectable in blood. Follow-up strongly relies on examination of the CSF. Due to the painful lumbar punctures, patients stay away from post-treatment controls unless serious disease symptoms appear [3-5]. Few precise guidelines can be found for decision making during follow-up if parasites cannot be demonstrated. Occurrence of clinical symptoms gives a good indication of relapse but is no absolute proof, serological tests on blood are of no value due to the persistence of antibodies up to several years after successful treatment and detection of trypanosomes relies on insensitive techniques.

There is clearly an urgent need for improved serological tests, parasitological tests, and better tests for staging and post-treatment follow-up of HAT. Examples of fields of research that may lead to new diagnostics are discussed below.

SEROLOGICAL TESTS: THE FUTURE

Improvements in serological diagnosis of HAT have to be expected at 4 levels: i) improved test formats; ii) identification of new antigens for antibody detection; iii) identification of disease biomarkers; iv) development of accurate antigen detection tests.

Improved test formats.

With a decreasing prevalence of HAT [6], cost-efficacy of active screening becomes low, leading to a progressive discontinuation of mobile team activities. The importance of passive detection in fixed health centers and the need for individual diagnostic tests will grow. Especially lateral flow tests and dri-dot agglutination tests are suited for such purposes. Examples of individual tests already in use for other neglected diseases in the developing world include the rk39 lateral flow test and LeptoTek Dri-Dot test for diagnosis of visceral leishmaniasis and human leptospirosis respectively. The rk39 lateral flow tests are easy to perform, accurate and reproducible [7]. Integration of the rk39 lateral flow test into field diagnostic algorithms of visceral leishmaniasis is envisaged [8]. For rapid serodiagnosis of human leptospirosis, the LeptoTek Dri-Dot [9] is a latex card agglutination test, comparable to LATEX/*T.b. gambiense* [10]. In LeptoTek Dri-Dot the reagent has been spotted and dried onto the agglutination cards. Cards are wrapped individually and as a result, a single assay test is obtained that can be stored at ambient temperature. The same assay format has been applied to develop tests for brucellosis and typhoid fever [11, 12] and could be adopted for LATEX/*T.b. gambiense*.

For epidemiological purposes, screening of large populations and repeated sampling, non-invasive tests may offer the advantages of high acceptability for participating subjects, low requirement for technical and medical support and avoidance of ethical problems [13-16]. Saliva testing can represent a non-invasive alternative for specific antibody detection. The principle has been explored for antibody detection in HIV, hepatitis, measles or rubella infection [15, 17, 18], and in parasitic infections such as schistosomiasis and visceral leishmaniasis [19, 20]. Also for diagnosis of HAT, saliva testing may represent a non-invasive alternative for antibody detection [21]. The use of urine has never been reported for diagnosis of human African trypanosomiasis. For diagnosis of visceral leishmaniasis however, an experimental card agglutination test detecting leishmania antigens in urine has been developed [22, 23].

Identification of new antigens for antibody detection.

New antibody detection tests should be simpler, more sensitive and more specific. The latter can be achieved by using purified, recombinant or synthetic antigens rather than whole organisms.

A possible strategy to make recombinant antigens is to choose a candidate protein from literature or *de novo* research. The search for serodiagnostic markers may be facilitated by the availability of the kinetoplastid genome sequences. In addition, many parasite-specific proteins with diagnostic potential have been identified as a by-product of fundamental research on African trypanosomes. Important considerations for the choice of a diagnostic antigen are [24]: i) the selection of a protein that is invariant. For example, an antigen to be used for diagnosis of both *T.b. rhodesiense* and *gambiense* should preferentially occur in both subspecies, or be present in high similarity. ii) Identification of a trypanosome unique protein with no homologue in other genera reduces the risk of cross-reactivity. Knowledge of the proteome permits searches for homology in other species and subspecies allowing selection on specificity. iii) The induction of a strong immune response to the protein in infected individuals.

The first 2 steps can be done *in silico*. Once a candidate has been identified, its presence should be experimentally confirmed in isolates of the *Trypanozoon* group for example by RT-PCR. A next step is to clone the corresponding gene and to sequence it. The corresponding recombinant protein or peptide can now be expressed. A variety of *in vitro* expression systems, which all have their advantages and disadvantages, is available, including bacteria, insect and yeast cells. If successful expression is obtained, and the recombinant protein is sufficiently purified, it can again be tested for its diagnostic potential. Some candidate antigens include [24] invariable surface glycoproteins [25, 26], heat shock proteins [27], proteases and their inhibitors [28-30], serum resistance associated protein [31], flagellar pocket glycoprotein [32, 33], cytoskeleton proteins [34], etc.

For the first time, scientists and laboratories with such candidate antigens are collaborating in screening for the potential of different antigens for serodiagnosis of both *T.b. gambiense* and *T.b. rhodesiense* disease. A total 32 antigens have been collected, most recombinant, and screened by dotblot and ELISA for their diagnostic potential by a private company using a collection of HAT and non-HAT control sera originating from a.o. Congo, Uganda, Benin and Europe. The most promising antigens are retained for further evaluation and incorporation in point-of-care tests at a later stage.

An alternative approach is the use of short synthetic peptide antigens, which can be identified based on the amino acid sequence of known proteins with diagnostic value or through panning of random peptide phage display libraries.

The first approach has been applied on LiTat 1.3 VSG, the diagnostic antigen in use in the CATT reagent, and on invariable surface glycoprotein 75. Peptides were derived from the amino-acid sequence of these proteins but were insufficiently specific and sensitive for diagnosis (P. Büscher, unpublished results). This result can be explained by presence of conformational and thus non-linear epitopes a.o. formed by the N-terminal domain of VSG. The phage display technology can be used to identify "mimotope" peptides, mimicking conformational or non-protein carbohydrate epitopes. Through insertion of DNA in the sequence of surface proteins of bacteriophages (viruses infecting bacteria such as *Escherichia coli*), the corresponding peptide is expressed as a fusion protein on the surface of the phage. Random phage display libraries, expressing random peptides, are commercially available. Selection of mimotope peptides from these libraries is obtained through affinity binding with specific antibodies after which the DNA sequence of the insert is determined. The amino acid sequence of the corresponding peptide is deduced and the peptide can be made synthetically. This technique has been used to identify mimotopic peptides for a.o. Lyme disease, hepatitis C, typhus, and tuberculosis [35-38]. We are currently trying to identify mimotope peptides for VSG antigens using specific monoclonal antibodies and HAT patient antibodies.

Identification of disease biomarkers.

Biomarker identification is based on the hypothesis that a disease state, including HAT, is characterized by a change in protein profile. Thus, diseased and non-diseased persons, such as HAT and non-HAT infected persons would have different protein profiles in their body fluids, tissue samples and so on. Protein profiles are studied by proteomic fingerprinting techniques, such as two dimensional gel electrophoresis or surface enhanced laser desorption-ionization time of flight mass spectrometry or SELDI-TOF. The later has been applied for discovery of biomarkers for HAT [39]. The technique consists of a pre-fractionation on protein chip arrays, which act as a chromatographic surface. It is followed by a laser desorption step by which proteins are released from the surface, and time of flight mass spectrometry causing separation of proteins according to mass/charge (protein ions of different masses fly at different velocities). The obtained spectra of diseased (HAT) and non-diseased persons are compared for up- or down-regulated proteins, which may act as biomarkers for the disease. SELDI-TOF claims to be a fast, high throughput alternative for 2-dimensional electrophoresis allowing high accuracy biomarker detection without pre-supposition on marker identity. Although the technique has potential, SELDI-TOF has been criticized [40]. It has been extensively applied in cancer research where different research groups have discovered different biomarkers, sometimes without any relationship or overlap. Results have hardly been validated with non-SELDI methods and efforts to identify biomarkers have remained minimal. Due to the highly specialized, expensive equipment and the extensive expertise needed, the technique cannot by readily applied in the field for diagnosis of HAT. Further identification of differentially expressed biomarkers is necessary, and difficult. Only then, findings can be translated into non-SELDI based tests (ELISA, immunoblotting, agglutination, lateral flow) which can be further be evaluated for their diagnostic potential for HAT.

Development of new, accurate antigen detection tests.

A diagnostic test based on the detection of antigen, would have the potential of detecting cases with active disease only and would be useful for follow-up after treatment. Theoretically, if they are able to detect the low antigen concentrations present, antigen tests may eliminate the need for laborious and insensitive parasitological examinations. Unfortunately, the antigen detection tests developed so far [41-44] have not been validated and are not recommended for diagnostic purposes.

The feasibility of developing new antigen detection tests is presently being assessed. In order to detect parasite antigens, camel heavy-chain antibodies (nanobodies) [45] and RNA aptamers [46] are being evaluated. These structures, unlike classical antibodies, are able to penetrate into the VSG surface coat of trypanosomes where they recognize carbohydrates or conserved protein moieties of the VSG, common for all Salivaria trypanosomes.

An alternative approach is to use the single chain variable fragment (scFv) antibody engineering technology in the development of optimized antibody probes for trypanosome antigens in blood. In this way, high-affinity antibody fragments for a number of T. brucei proteins can be generated, from which those that are best suited for antigen detection in human samples are selected.

PARASITOLOGICAL DIAGNOSIS: THE FUTURE

As long as antibody detection has a limited specificity and antigen tests remain to be developed and evaluated, the need for parasite detection will persist. mAECT, the most sensitive technique actually available, with a detection limit around 50 trypanosomes/ml is still not able to detect parasites in all patients. New parasitological techniques need to be at least equally sensitive as mAECT and simpler to perform. Novelties to be expected in trypanosome detection include better visualization and/or new or improved concentration techniques.

The availability of low cost fluorescence microscopes opens opportunities for better visualization of the trypanosomes using fluorescent markers. Fluorescent labeling of trypanosomes can be achieved by using fluorescently labeled specific antibodies [47] or other trypanosome binding structures carrying a fluorescent label (Fig 1).

Peptide nucleic acid (PNA) probes are pseudopeptides consisting of a polyamide formed by repetitive units of N-2-aminoethyl glycine units to which nucleobases are covalently attached. PNA probes hybridize to complementary nucleic acid targets and PNA fluorescence in situ hybridization (FISH) has been used to target *Trypanozoon* multiple copy 18S rRNA [48], greatly facilitating the detection of a single trypanosome. Major limitations that may limit practical implementation of these techniques might include the need of a special fluorescence microscope and the limited stability of the reagent.

New techniques for concentration of trypanosomes make use of trypanosome binding antibodies, nanobodies and/or RNA aptamers. After their labeling with magnetic beads, the antibodies/nanobodies/aptamers can be mixed with a suspect's blood and will bind to the parasite if present. They are then recovered using a magnet, together with the attached parasites [49], which can be observed under a microscope. Other approaches include separation of parasites using cellulose membrane filters, separation of parasite from blood in an electric field, density separation, using other gels, biosensors, etc.

MOLECULAR DIAGNOSIS: THE FUTURE

Possible solutions to address the limited sensitivity of actual parasite detection techniques include the detection of trypanosome nucleic acid from a patient's sample, which can be considered as a surrogate for parasite detection. At present molecular diagnosis is too cumbersome to be applied in health centers with minimal equipment and experience in molecular biology. Field applicability of nucleic acid detection could be increased in several ways, including eliminating the need for a thermocyclers by performing the nucleic acid amplification isothermally or eliminating the cumbersome methods of amplicon detection.

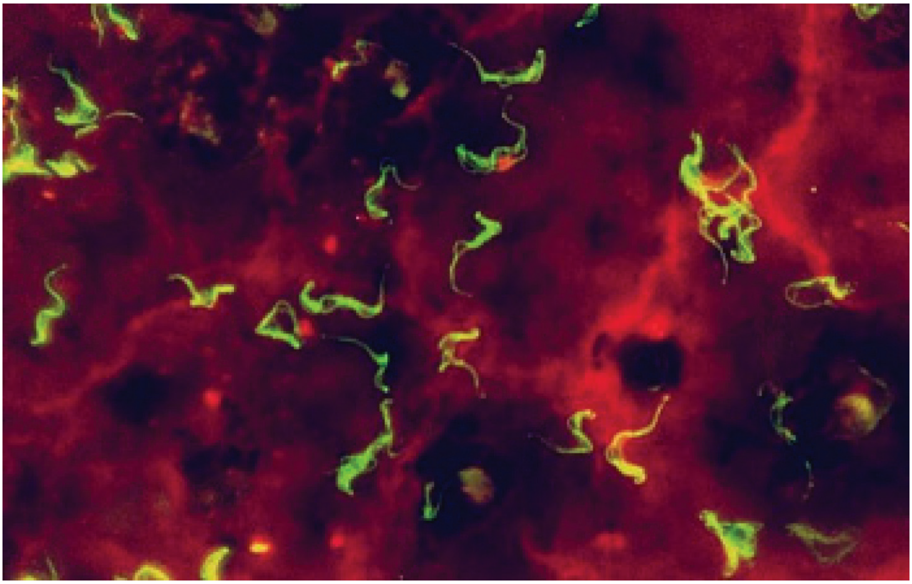

Fig. 1a

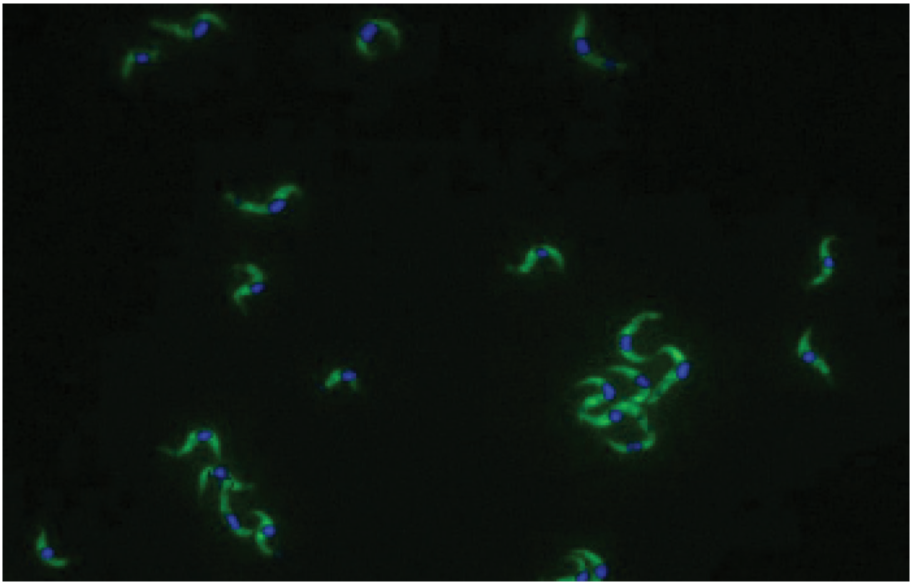

Fig. 1b

Figure 1 - Improved visualization of trypanosomes in a thick drop preparation using (a) uorescently labeled trypanosome specic polyclonal antibodies or (b) PNA probes in FISH, counterstained with DAPI to visualize the nucleus.

Simplified amplicon detection.

PCR amplicons are usually detected using UV transillumination after being electrophoresed in the presence of ethidium bromide which is a carcinogen. Oligochromatography provides a dipstick format for detection of amplified PCR products, which are visualized by hybridization with a gold-conjugated probe (Fig. 2).

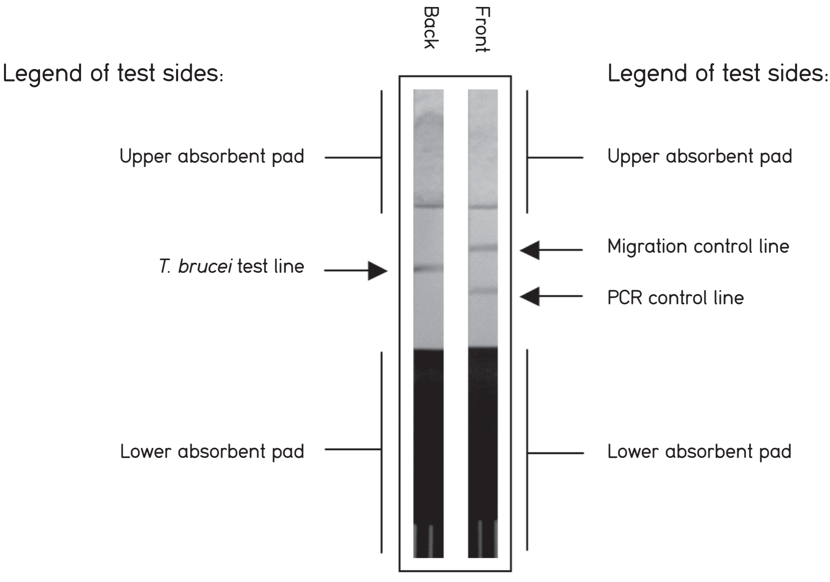

Figure 2 - PCR-oligochromatography test result from a sample containing *T. brucei* DNA [50].

Applied onto HAT patient samples, oligochromatography combines the sensitivity and specificity of PCR with the simplicity and speed of membrane chromatography (result after 5 minutes) [50]. It doesn't require sophisticated equipment since it is performed at constant temperature and has clear potential for implementation as a reference diagnostic test in midlevel-equipped laboratories in endemic countries. Further refinements may include its combination with nucleic acid amplification at constant temperature such as nucleic acid based sequence amplification (NASBA) or loop-mediated isothermal amplification (LAMP).

Isothermal nucleic acid multiplication.

The NASBA technology utilizes 3 enzymes (reverse transcriptase, RNase H and T7 RNA polymerase) and target specific oligonucleotides, of which one encodes the T7 RNA polymerase promoter (for more information and an animation on the principle of NASBA see www.ibi.cc/nasba%20 step%20by%20step.htm). The entire NASBA process is conducted at 41°C, and generates single-stranded RNA as the end product. The typical level of amplification is at least a factor of 10^9. NASBA for detection of *T.b. gambiense* and *T.b. rhodesiense* in blood and cerebrospinal fluid using the rather sophisticated IQ5 real time analysis for detection of the amplicons has been performed with success (Mugasa et al., personal communication), and it's adaptation into a NASBA-oligochromatographic test is ongoing (G. Schoone, personal communication).

Loop mediated isothermal amplification of DNA or LAMP is a DNA amplification method that uses 4-6 different primers specifically designed to recognize 6 distinct regions on the target gene. The reaction process proceeds at a constant temperature using a strand displacement reaction (for more information and an animation on the principle of the LAMP method see loopamp.eiken.co.jp/e/lamp/). Amplification and detection of a gene can be completed in a single step, by incubating the mixture of samples, primers, DNA polymerase with strand displacement activity and substrates at a constant temperature (about 65°C). It provides high amplification efficiency, with DNA being amplified 10^9-10^{10} times in 15-60 minutes. Because of its high specificity, the presence of amplified product can indicate the presence of target gene. The result can be read visually, either because of a white precipitation byproduct which is formed by reaction of magnesium ions with the pyrophosphate ions produced during amplification, or by addition of SYBR green I which is a DNA intercalating dye and turns from orange to green in the presence of large amounts of DNA (Fig. 3).

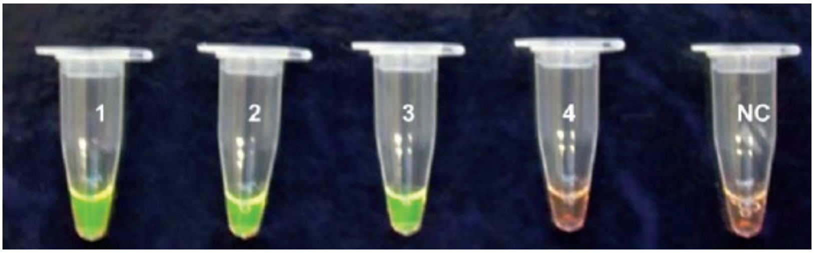

Figure 3 - Visual Appearance of LAMP results after the addition of SYBR Green I [51].
Tubes 1-3: Positive samples produce a green color almost immediately.
Tubes 4-5: Negatives remain orange.

LAMP has been successfully applied to detect trypanosome DNA specific to the subgenus *Trypanozoon*, *T.b. rhodesiense* and *T.b. gambiense*, depending on the primers used [51-53]. Overall, LAMP tends to be more sensitive than the corresponding PCRs. The test has great potential for field application, as has been shown for diagnosis of tuberculosis [54], but remains to be evaluated on a larger scale on HAT clinical samples and might need some further simplifications.

STAGE DETERMINATION AND FOLLOW-UP: THE FUTURE

Research on improvement of stage determination and post-treatment follow-up is being conducted on different levels: i) Incremental improvement of known markers (including improvement of existing techniques and their translation into point-of-care tests); ii) validation of –previously discovered-biomarkers; iii) discovery of new markers.
Identification of non-invasive markers for the disease stage and for follow-up has been set as a priority for research but seems difficult to accomplish.

Improving existing techniques.

Improvement of parasite detection in CSF can be obtained using one of the techniques already mentioned above.

Improved methods for cell counting have been studied for diagnosis of meningitis, but cannot directly be applied for staging of HAT. Once introduced, disposable counting chambers are usually preferred above traditional counting chambers, and increase standardization, save time and allow easy counting [55]. Based on the findings that a white blood cell increase in CSF of HAT patients is a B-cell increase [56, 57], one could label these cells using an anti-CD19 or anti-CD20 fluorescent conjugate. This allows B-cell detection by flow cytometry or with a fluorescent microscope. Although the former is sensitive, the cost of a machine and the reagents in combination with the need for extensive maintenance of the machine are limiting factors. Hematology analyzers require big sample volumes and are usually not reliable when less than 30 white blood cells/µl are present or the majority of cells are lymphocytes which is the case in HAT [58-62]. The use of urine reagent strips for easy estimation of white blood cell count has been used with variable success in meningitis, but is not applicable in HAT due to the relatively low cell counts [63, 64] and the absence of polymorphonuclear cells expressing leukocyte esterase, needed for visualization on such strips.
However, for follow-up, improvement of parasite detection and better, uniform, well validated guidelines for the white blood cell count are actually missing but may already improve diagnosis of cure and relapse in many patients.

Efforts to transform promising alternative tests for disease staging in more user friendly test formats are ongoing. LATEX/IgM [65] is being transformed into individual tests using the dridot technology already discussed. ELISA assays to measure auto-antibodies against galactocerebrosides and neurofilament have been transformed into dot-ELISAs [66] but may need further simplification.

Validation research.

Some biomarkers have been described from animal models, in other neurological diseases or an increase can be predicted based on the pathogenesis of HAT. They remain to be validated on larger groups of samples before transformation into point-of-care tests and their introduction into the field.

Inflammatory events in the central nervous system may be detected by measuring components of the B or T cell response in CSF. B-cell products may include various locally produced antibodies in response to brain infection [67-69], or immune complexes [70]. T-cell products may origin from microglia, astrocytes or other brain cells. Neopterin [71], prostaglandin D2 [72], various cytokines such as IL-1α, IL-6, IL-8, IL-10 [71, 73-75] and the chemokines CXCL-8, CCL-2 and CCL-3 [75] have been detected in HAT. The availability of bead assays or micro-arrays allows rapid analysis of different cytokines in parallel on small sample volumes and might provide additional information about the cytokine profile in CSF of HAT patients. All these parameters mainly indicate the occurrence of inflammatory reactions and are not necessarily sleeping sickness specific nor brain specific since cytokines may end-up in CSF by diffusion from blood.

Brain specific or destruction markers detected in blood and/or CSF of HAT patients include glial fibrillary acidic protein and neurofilament in CSF [76, 77]. Other markers are being explored. Auto-antibodies against brain components such as myelin [78], galactocerebrosides [79] or neurofilament [80] have also been detected in HAT. Such brain specific antibodies may be induced by antigens leaking out of the central nervous system but could also result from the polyclonal B-cell activation, or an antibody response against parasite epitopes cross-reacting with brain specific antigens. As for inflammation related parameters, brain specific markers are not exclusively linked to trypanosomiasis.

An exciting field of research examines parameters related to the sleep disturbances observed in HAT and may lead to non-invasive tests for staging or follow-up. Daytime sleepiness is a symptom occurring frequently during meningo-encephalitic stage HAT. Sleep and wakefulness can be recorded by polysomnography, which is able to objectively differentiate between wakefulness, REM (rapid eye movement) sleep and non-REM sleep and its four stages. Using polysomnography, two major disturbances have been recorded in meningo-encephalitic stage patients. Besides an alteration of the sleep-wake cycle with short sleep episodes that occur equally day and night, an alteration in structure of sleep episodes with frequent sleep onset REM sleep periods (SOREMP) is observed [81]. As such, sleep wake recordings may constitute a useful tool for diagnosing central nervous system involvement which may occur early, when classical CSF parameters are still normal [82]. After treatment, sleep disturbances should improve, and their persistence may indicate that cure is delayed or that neurological sequellae are present.

Discovery of staging biomarkers.

Similar to disease biomarkers, discovery research using protein profiling by proteomic fingerprinting techniques may also lead to the discovery of new biomarkers for the disease stage. Protein profiling of CSF through two dimensional gel electrophoresis has been successfully applied for discovery of new markers potentially relevant for neuro-degeneration and stroke [83], and is now being applied on cerebrospinal fluid samples of HAT patients.

TOWARDS FUTURE DIAGNOSTICS: REQUIREMENTS AND CHALLENGES

The final result of the research described above as well as other HAT diagnostic research, should be a diagnostic test appropriate for use in HAT foci. WHO has developed a list of general characteristics, abbreviated using the acronym ASSURED, that make an appropriate diagnostic for resource–limited settings [84]. Such diagnostics should be **a**ffordable for those at risk of infection, **s**ensitive, **s**pecific, **u**ser-friendly, **r**apid and robust, **e**quipment free and **d**elivered to those in need. Especially challenging for HAT is that most of the diagnostic activities are performed at the lowest level of laboratory infrastructure, where no or only minimal laboratory infrastructure are available, implying the lack of electricity or clean water, lack of cold chain, minimal training, while a rapid answer is required [85]. These conditions greatly limit the use of existing tests since, if necessary, purified water will have to be provided and electricity should be supplied in the form of batteries. The lack of a cold chain limits the sample storage time while reagents should be stable to resist the fluctuating temperatures.

Besides the level of infrastructure in which tests will be used, additional barriers currently hurdling test development are shown in figure 4. Although this is a general flow chart, many if not most of the barriers are applicable to HAT and many of the tests and markers discussed above might not make it to the end of the pipeline.

NEW INITIATIVES

Due to the lack of a commercial market, diagnostic test development in the past has been mainly driven by academic research, which, taking into account the above mentioned barriers, is one of the many reasons why so few diagnostics for HAT have made it. More and more, public-private partnerships are built which should increase chances to overcome barriers for diagnostic test development, including HAT diagnostics. Such a new initiative is the Foundation of Innovative New Diagnostics, FIND (www.finddiagnostics.org), dedicated to the development of rapid, accurate and affordable diagnostic tests through public-private partnerships. FIND's specific objectives are: i) to develop, in partnership with academia, public and private research institutes and industry, new diagnostic approaches that have been proven in principle; ii) to transform them into effective products for identifying poverty-related diseases and; iii) to compare and evaluate these products in coordinated laboratory and field trials as part of FIND's milestone process from customer requirement to access to cheap and effective diagnosis.
Through FIND's efforts, future diagnostics may therefore be helped to overcome the barriers discussed above.

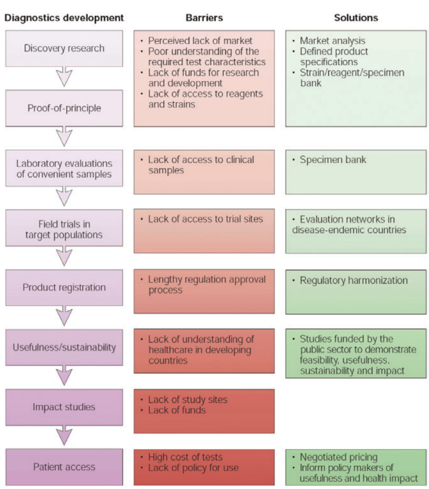

Figure 4 - An overview of the different steps in the development of a diagnostic from discovery research to test use, possible barriers in development and solution [84].

Early 2006, FIND, jointly with the WHO, has launched a HAT diagnostics program supporting some diagnostic developments. The areas of emphasis include i) improvement in parasite separation from blood and CSF (www.finddiagnostics.org/activities/ss/parasite_detection.shtml) ii) serological tests (www.finddiagnostics.org/activities/ss/serological_diagnosis.shtml), iii) molecular tests (www.finddiagnostics.org/activities/ss/molecular_diagnosis.shtml), iv) staging of the disease (www.finddiagnostics.org/activities/ss/staging.shtml), v) and establishment of a specimen bank facilitating diagnostic research (www.finddiagnostics.org/activities/ss/ specimen_bank.shtml). These projects should enable the development of user-friendly diagnostic tests and progress well.

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